



V3 SMALL SCALE

SIMULATION REVIEW AND CAD CORRECTIONS

ITEM	DETAIL
Document	V3-SIM-2026-09-29, revision A, 29 September 2026
Design reviewed	Fusion 360: ATLAS / V3 / SMALL_SCALE_V3, version 34
Method	Airframe Designer flight simulator with the PX4 v1.17 flight controller in the loop, 154 simulated flights
Prepared for	CAD design, to apply the corrections in section 4
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1 SUMMARY

We read the current CAD model, worked out the weight and balance of the aircraft from the part labels, and flew it in the simulator. The aircraft **flies as drawn**: all 9 test hovers were stable and none crashed. Two things hold it back, and both are simple CAD changes.

- **The batteries sit too far forward.** The balance point (centre of gravity, CG) is 50 mm ahead of where the nine fans share the work evenly. The nose fans work harder than the rest and the aircraft wanders more in a hands-off hover.
- **The optical flow sensor (H-FLOW) points the wrong way in flight.** The aircraft hovers 20° nose-up, so a sensor mounted flat looks 20° off vertical. That makes the indoor position hold less accurate.

71 %

less wander in a hands-off hover after moving the batteries (1.86 to 0.53 m)

47 %

load on the busiest fan after the change, down from 55 %

26 %

less drift when holding position on the optical sensor, after tilting its mount

Required CAD changes

1. **C1:** move the rear block of four batteries 195 mm towards the tail, and lower the two outer ones 10 mm.
2. **C2:** tilt the H-FLOW mount 20.4° so it looks straight down while hovering.
3. **C3:** correct the weight labels on the parts (missing labels, duplicates, one inconsistent pair).
4. **No change:** jetfoil angles and nose fans stay as drawn.

2 BACKGROUND AND METHOD

The simulator is a digital copy of the aircraft: its geometry and weights, the thrust of every fan, the jet turned by the jetfoils, the legs on the ground. It is connected to the same flight-control software that runs on the real aircraft (PX4), which flies it exactly as it would fly the real one. The work followed five steps:

1. **Read the CAD.** The open Fusion design was read directly: every part, its position and its shape. The angle at which each jetfoil releases its jet was measured on the model surface: 69.0° (inner), 59.3° (middle) and 49.4° (outer) below the fan axis.
2. **Weigh it.** Only parts whose name carries a weight label (for example **battery_5200mAh_M760g** = 760 g) were counted. Parts without a label, and unlabelled copies that sit exactly on top of a labelled part, count as zero.
3. **Find the balance point.** From the weights and positions we computed the CG and, for each battery position, how hard each fan must work to hover.
4. **Fly it.** Each design was flown hands-off: take off, hover 12 s, land. Every case was flown three times with different sensor noise, and at its hover angle and 0.5° either side, so a result does not depend on one lucky flight.
5. **Test the optical sensor.** A model of the H-FLOW (camera that tracks the floor plus a distance sensor) was added, GPS was switched off, and the aircraft was asked to hold its position indoors.

3 FINDINGS

3.1 WEIGHT AND BALANCE

The labelled parts add up to **11.83 kg**. The six 5200 mAh batteries are the heaviest group (4.56 kg, 39 %), so where they sit decides the balance.

GROUP	COUNT	MASS
XFLY 80 mm fans	9	2.97 kg
Batteries 5200 mAh	6	4.56 kg
Jetfoils (920 g + 800 g)	2	1.72 kg
ESCs and power boards	3 + 3	1.13 kg
Carbon tubes	8	0.41 kg
Nose carrier, jetfoil mounts (left side only)	1 + 2	0.68 kg
Avionics bay, H-FLOW, 1600 mAh pack	3	0.37 kg
Total (labelled parts only)		11.83 kg

As drawn, the CG sits at the geometric middle of the airframe. That is not where it should be for this aircraft. The jetfoils push along the jets leaving their trailing edges, which are far back and low, so the point over which the fans share the load evenly lies about **50 mm further back**. With the CG there, the busiest fan drops from 55 % to 47 % of full power and the least loaded one rises from 31 % to 35 %: more even, more reserve.

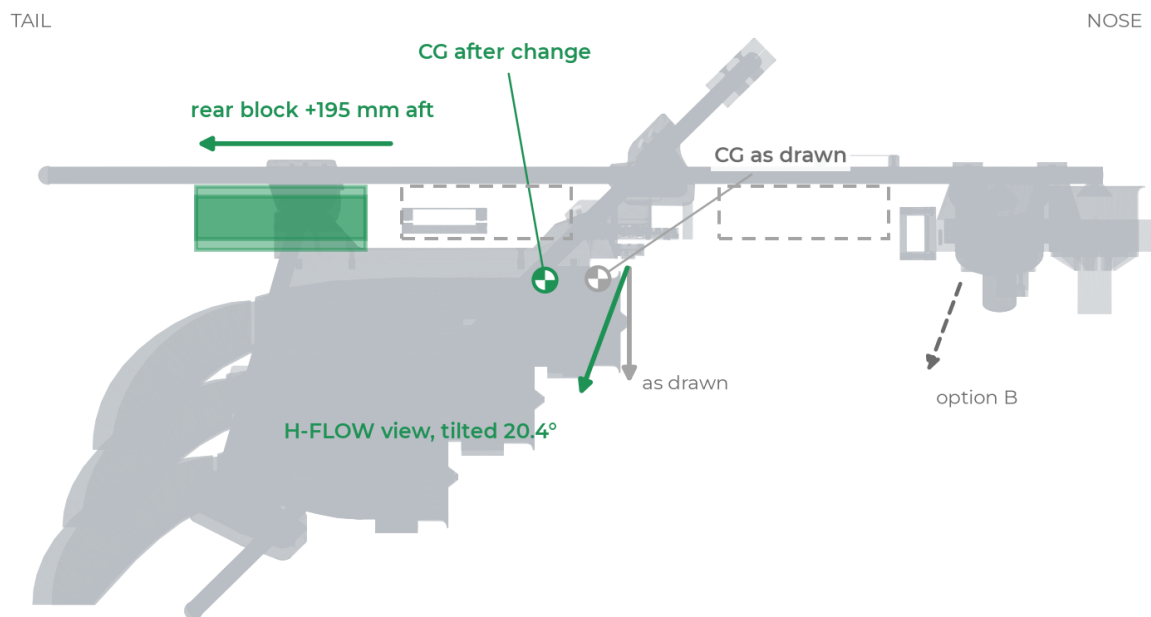


Figure 1. Side view, nose right. Grey dashed: batteries and CG as drawn. Green: after change C1, and the H-FLOW view direction after change C2 (20.4° tilt).

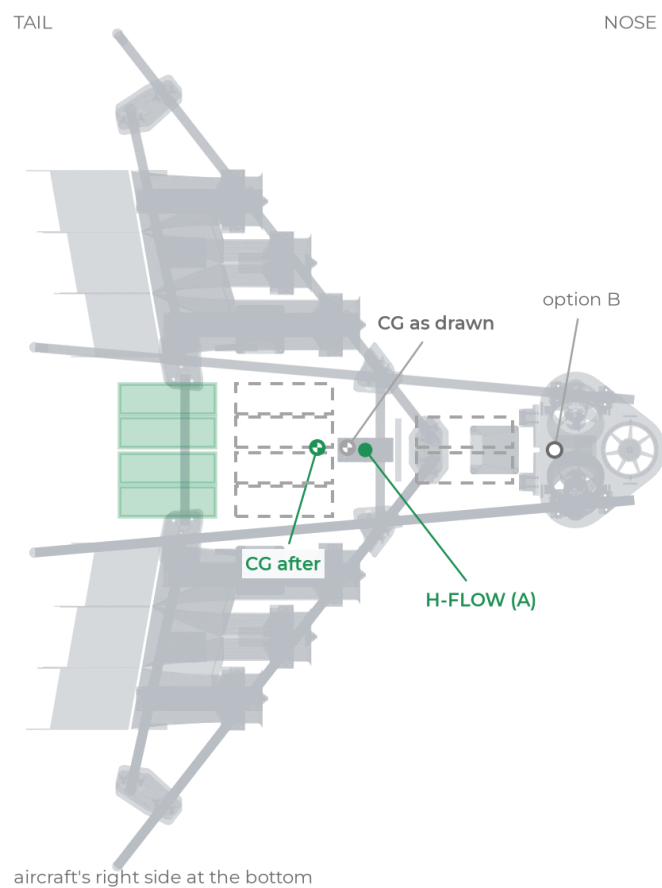


Figure 2. Top view, nose right. The two front batteries stay; the rear block of four moves towards the tail.

3.2 STABILITY IN HOVER

Several battery layouts were flown. "Drift" is how far the aircraft wanders during a 12 s hover with nobody touching the sticks: a direct measure of how settled it is.

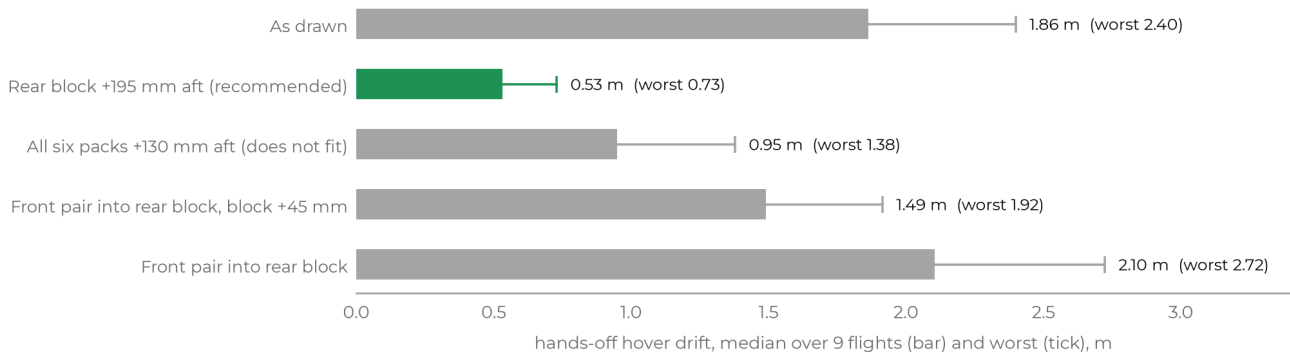


Figure 3. Hover drift for each battery layout. Every layout flew all 9 of its flights without a crash.

BATTERY LAYOUT	FLIGHTS OK	DRIFT MEDIAN / WORST	PITCH HOLD ERROR	BUSIEST FAN	TOUCHDOWN SPEED
As drawn	9/9	1.86 / 2.40 m	0.34°	55 %	0.43 m/s
Rear block +195 mm aft (recommended)	9/9	0.53 / 0.73 m	0.05°	50 %	0.38 m/s
All six packs +130 mm aft (does not fit)	9/9	0.95 / 1.38 m	0.15°	49 %	0.38 m/s
Front pair into rear block, block +45 mm	9/9	1.49 / 1.92 m	0.24°	50 %	0.40 m/s
Front pair into rear block	9/9	2.10 / 2.72 m	0.35°	49 %	0.49 m/s

Moving all six batteries back together would work as well, but the front pair would then sit inside the flight controller. Moving only the rear block gives the best result and fits.

3.3 CONTROL AUTHORITY

Authority is the turning force the fans have in reserve while they hold the aircraft up. It decides how firmly the aircraft can correct itself and follow the pilot.

AXIS	RESERVE MOMENT	AS ACCELERATION	FLOWN TEST RESULT
Roll (tilt left/right)	±10.5 N·m	915 °/s ²	reaches the commanded 6° in under 2 s
Pitch (nose up/down)	±20.0 N·m	1237 °/s ²	reaches the commanded 6° in under 2 s
Yaw (turn on the spot)	±3.3 N·m	155 °/s ²	turns 38 °/s right, 33 °/s left of 60 °/s asked
4 m/s side wind			attitude held within 2.5°; heading swings 28 to 31°

Roll and pitch have plenty in reserve. **Yaw is the weak axis**, about six times weaker than roll, because all nine fans spin the same way. The aircraft weathervanes in a side wind. This is a known design limit (counter-rotating fan pairs would remove it) and is outside the scope of this report; the battery change does not make it worse.

3.4 OPTICAL FLOW SENSOR (H-FLOW)

Indoors there is no GPS, so the aircraft holds its position by watching the floor with the H-FLOW camera and measuring its height with the sensor's rangefinder. The flight controller assumes the sensor looks straight down while hovering. The aircraft hovers 20.4° nose-up, so the mount must be tilted by the same angle.

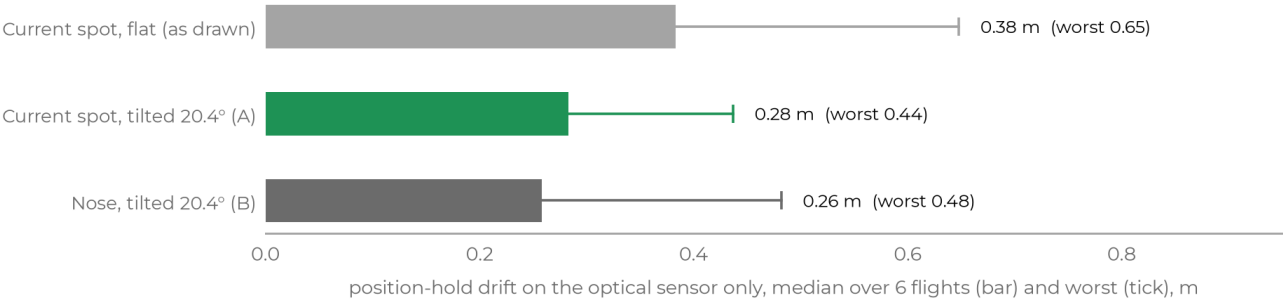


Figure 4. Position-hold drift using only the optical sensor (20 s hands-off hold, then a stick push and stop). For comparison, the same hold on GPS drifts 0.07 m.

MOUNT	DRIFT MEDIAN / WORST	HEIGHT NOISE	FLIGHTS
Current spot, flat (as drawn)	0.38 / 0.65 m	1.4 cm	6
Current spot, tilted 20.4° (option A, recommended)	0.28 / 0.44 m	0.6 cm	6
Under the nose, tilted 20.4° (option B)	0.26 / 0.48 m	0.6 cm	6

Tilting the mount is what matters: it cuts the drift by 26 % and halves the height noise. Where the sensor sits (under the avionics bay or under the nose) makes no measurable difference in the simulator. One practical difference remains: the jets. Their air stirs up dust where it hits the floor, and dust in the camera's view can confuse it.

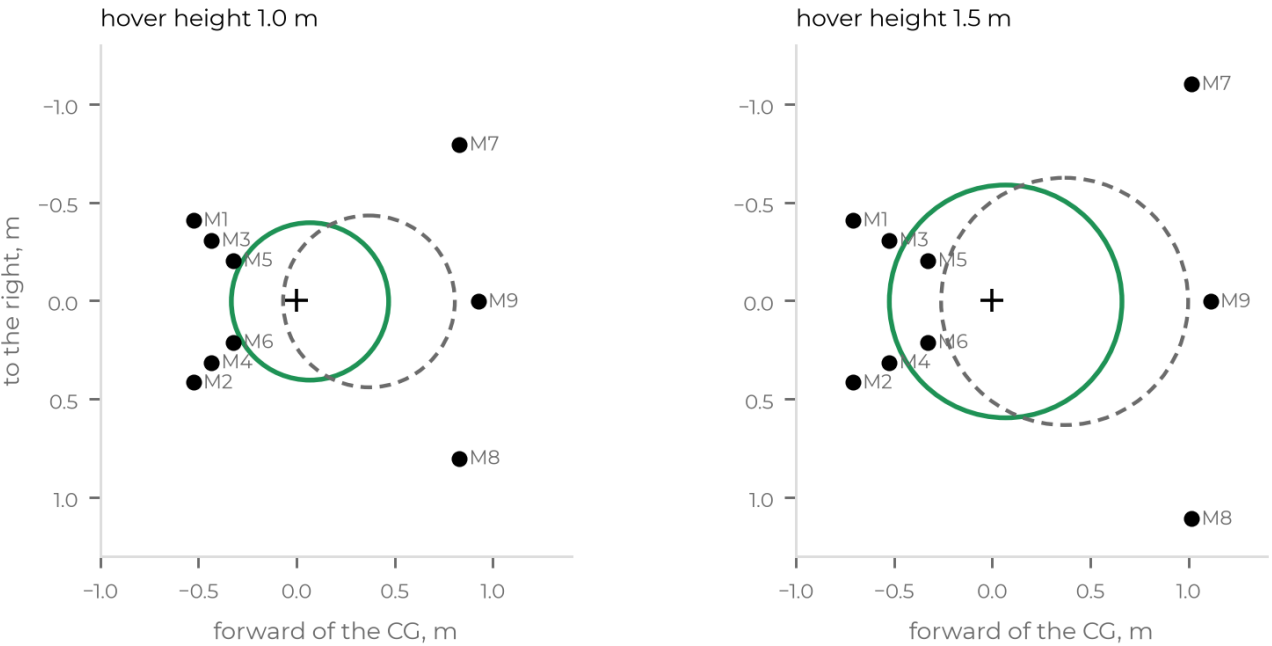


Figure 5. The floor seen from above while hovering. Black dots: where each fan's jet reaches the floor. Green circle: what the H-FLOW sees from its current spot. Grey dashed circle: from under the nose. The current spot keeps all jets out of view up to about 1.1 m hover height; the nose spot up to about 1.8 m.

4 REQUIRED CAD CHANGES

All positions are in the Fusion design coordinates of the open model, in millimetres: **X** to the aircraft's right, **Y** up, **Z** towards the tail (the nose is at negative Z).

C1 BATTERY PLACEMENT

1. Select the four rear batteries (the ones at Z = +150 mm), listed below.
2. Move them together **+195 mm along Z** (towards the tail).
3. Move the two outer ones (X = ±85 mm) a further **-10 mm along Y**. Without this they touch the mount parts (All_mounts) near Z 285 to 342 mm.
4. Rework the battery tray or straps for the new position. Leave the two front batteries (Z = -150 mm) and the slide-in carrier unchanged.

PART (OCCURRENCE)	CENTRE NOW X, Y, Z	CENTRE AFTER	MOVE
battery_5200mAh__M760g:5	-85, 435, 150	-85, 425, 345	Z +195, Y -10
battery_5200mAh__M760g:3	-30, 435, 150	-30, 435, 345	Z +195
battery_5200mAh__M760g(Mirror):2	30, 435, 150	30, 435, 345	Z +195
battery_5200mAh__M760g(Mirror):3	85, 435, 150	85, 425, 345	Z +195, Y -10

	CG X, Y, Z (MM)	HOVER ANGLE	BUSIEST / LEAST LOADED FAN
As drawn	-4, 372, 45	18.4°	55 % / 31 %
After C1	-4, 371, 95	20.4°	47 % / 35 %

C2 H-FLOW MOUNT

1. **Option A (recommended): keep the current position** under the avionics bay, lens centre at about X 0, Y 400, Z 15 mm. Rotate the mount **20.4° about the X axis** so that the lens turns from pointing straight down (-Y) towards the tail (+Z). On the bench the lens then looks slightly backwards; in the hover it looks straight down.
2. **Option B, only if the aircraft will hover higher than about 1.1 m:** under the nose, lens centre at about X 0, Y 373, Z -300 mm, same 20.4° tilt. The nose ESC (ESC__M200g:2) sits directly above this spot: mount the sensor on fixed structure next to it, not on the removable slide-in carrier.
3. In both cases keep a clear 42° cone below the lens (no legs, straps or cables in view) and at least 80 mm between the lens and the floor when parked (currently 386 mm).
4. If a later revision changes the hover angle, the tilt changes with it: tilt = hover angle.

C3 WEIGHT LABELS

The simulator only counts labelled parts. Until these are fixed, the real aircraft is heavier than the model and slightly unbalanced to the left (4 mm).

PART	PROBLEM	ACTION
All_mounts (14 bodies, 909 cm ³)	No weight label; the largest part not counted	Weigh and add __M<grams>g
Four_feet_assembled	No weight label	Add label
Clamp_cap_1, Clamp_cap_3 and their mirrors	No weight label	Add label

PART	PROBLEM	ACTION
Open CASCADE STEP translator 7.9 1.9(Mirror) and 1.11(Mirror)	Right-hand jetfoil mounts without labels: 280 g missing on the right	Rename to Jetfoil_mount_1__M120g(Mirror) and Jetfoil_mount_2__M160g(Mirror)
JET_FOIL_V3__M920g and JET_FOIL_V3__M800g(Mirror)	Same shape, different weights: 120 g difference left to right	Weigh both and correct
16x1000mm_carbon_tube(Mirror) ×2, 16x500mm_carbon_tube(Mirror), XFLY 80mm EDF(Mirror), battery_5200mAh(Mirror) ×3, PDB_XT90PW-F_CASE(Mirror)	Unlabelled duplicates on top of labelled parts; a mass report from Fusion counts them twice (9 batteries instead of 6)	Delete, or exclude from the design
XFLY 80mm EDF_M330g(Mirror) ×4	Label written with a single underscore	Rename to __M330g

KEEP AS DRAWN

Jetfoil angles (69° / 59° / 49°) and the nose fans. A steeper inner jetfoil was tested (72° and 75°); it flew no better at 72° and wandered twice as much at 75°.

5 LIMITATIONS

- Fan thrust (36 N each) and fan torque are taken from the ATLAS_09B aircraft, not measured on V3.
- The jet is assumed to follow each jetfoil all the way to its trailing edge. A real jet may leave the surface earlier and turn less.
- Only labelled parts are weighed (change C3). After relabelling, the CG will move; the study should be rerun before building.
- The optical sensor model sees an ideal, textured floor. Dust, lighting and vibration of the lens are not simulated; the jet check in figure 5 is geometric only. The sensor's field of view is taken as 42°.
- The legs are the simulator's provisional ones, parked at the hover angle.

APPENDIX A TERMS USED

TERM	MEANING
CG (centre of gravity)	The balance point of the aircraft: where all its weight acts.
Hover angle (trim pitch)	The nose-up angle the aircraft holds when hovering, set by its geometry and CG (20.4° after C1).
Fan load	How hard a fan works in the hover, in % of its full power. Lower and more even means more reserve.
Drift	How far the aircraft wanders while hovering with nobody touching the sticks.
Pitch hold error	How far the nose angle strays from the one asked for, on average.
Authority	The turning force the fans can spare while holding the aircraft up.
Position hold	Flight mode in which the aircraft stays in place by itself; indoors it relies on the H-FLOW.
Weathervaning	The aircraft turning its nose into or away from the wind because it cannot resist the wind's twisting force.
SITL	Software in the loop: the real flight-control software flying the simulated aircraft.

APPENDIX B COORDINATES

The simulator uses x forward, y right, z down, with the origin on the centreline at Fusion Y = 450 mm. Conversion to Fusion (mm): Fusion X = 1000·y, Fusion Y = 450 - 1000·z, Fusion Z = -1000·x. All numbers in section 4 are already converted.